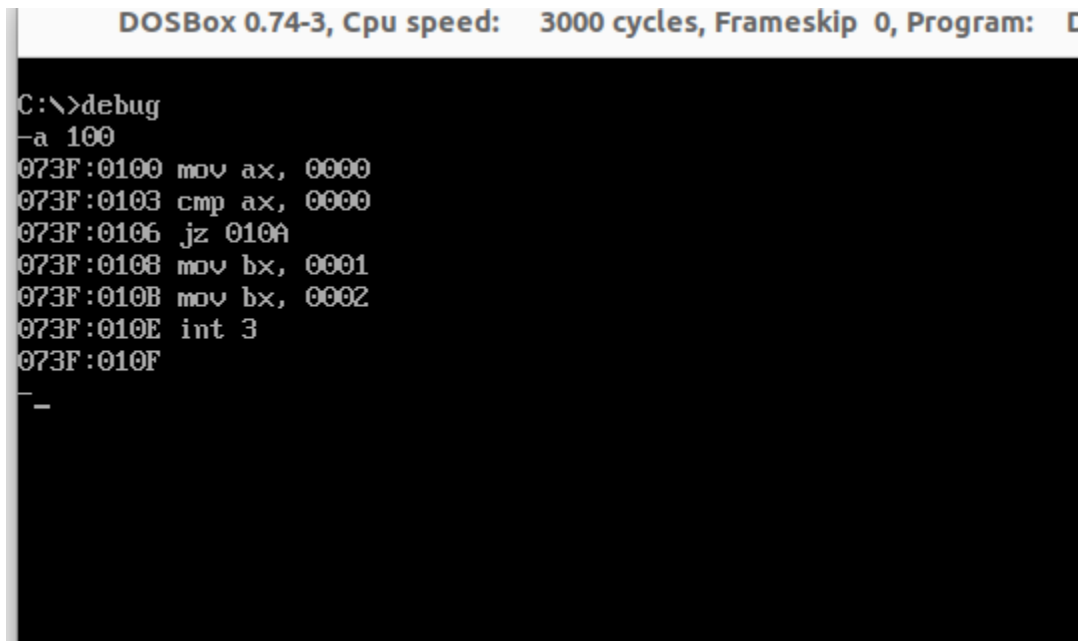


Make Logical Decisions using DOS DEBUG

Example: Checking if a number is zero using **JZ**

1. Open DEBUG:
2. Assemble a small decision-making routine at **100**:



```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: I
C:\>debug
-a 100
073F:0100 mov ax, 0000
073F:0103 cmp ax, 0000
073F:0106 jz 010A
073F:0108 mov bx, 0001
073F:010B mov bx, 0002
073F:010E int 3
073F:010F
-
```

3. Trace through the program using **t** to observe how the instruction pointer (**IP**) jumps past line **0108** directly to **010A** because the zero condition was met.

073F:010B mov bx, 0002

073F:010E int 3

073F:010F

-

-t

AX=0000 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000

DS=073F ES=073F SS=073F CS=073F IP=0103 NV UP EI PL NZ NA PO NC

073F:0103 3D0000 CMP AX,0000

-t

AX=0000 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000

DS=073F ES=073F SS=073F CS=073F IP=0106 NV UP EI PL ZR NA PE NC

073F:0106 7402 JZ 010A

-t

AX=0000 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000

DS=073F ES=073F SS=073F CS=073F IP=010A NV UP EI PL ZR NA PE NC

073F:010A 00BB0200 ADD [BP+DI+0002],BH SS:0002=3E

-t

AX=0000 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000

DS=073F ES=073F SS=073F CS=073F IP=010E NV UP EI PL NZ NA PO NC

073F:010E CC INT 3

-